`LakeTemperature` Testing Software

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Background

- We are testing a module of *E3SM,*[1] an environmental model. This module, `LakeTemperature`, is part of the greater *ELM* (*E3SM* Land Model) which was originally a fork of the *CLM* (Community Land Model).
- We are able to test specifically this module using *SPEL,*[2] a software that adds a *NetCDF*[3] interface for the constants, inputs, and outputs of a module of *E3SM* and compiles that specific part of the model. The executable produced by this is named `elmtest.exe`.

1. “Energy Exascale Earth System Model.” *E3SM*, U.S. Department of Energy: Office of Science, n.d., e3sm.org/.

2. Schwartz, Peter. “SPEL_OpenACC.” *GitHub*, 16 Oct. 2025, github.com/peterdschwartz/SPEL_OpenACC.git.

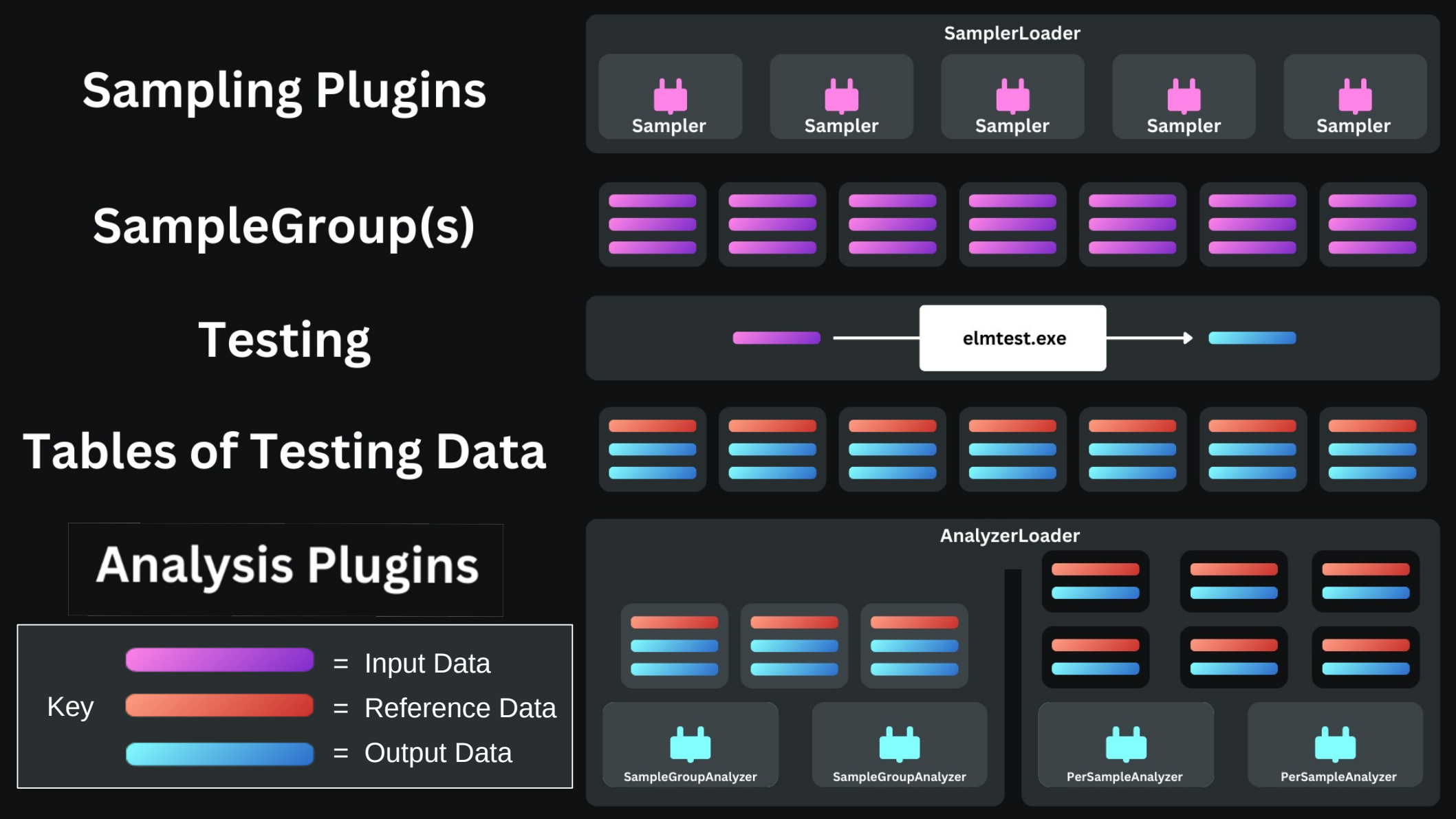
3. “NetCDF.” *NSF UNIDATA*, n.d., www.unidata.ucar.edu/software/netcdf.

Project Goals

- Develop a software that can systematically change the constants used in the `LakeTemperature` model and allow us to visualize and observe patterns in how this affects its output.
- Apply testing techniques to try and find values for constants that will cause physically impossible or otherwise erroneous outputs. Apply assertions pertaining to the values of constants, inputs, and outputs based around physical laws and other constraints provided by domain scientists.

Our Testing Solution

- Automates the process of modifying constants in `spel-constants0001.nc` and the process of analyzing the resulting values in `fut-outputs0001.nc`
- Both the process of choosing values for each constant and the process of analyzing the resulting outputs is extendable through sampling plugins and analysis plugins respectively.


Sampling Plugins
SamplerLoader
Sampler
Sampler
Sampler
Sampler
Sampler
SampleGroup(s)
Testing
elmtest.exe
Tables of Testing Data
Analysis Plugins
AnalyzerLoader
Key
= Input Data
= Reference Data
= Output Data
SampleGroupAnalyzer
SampleGroupAnalyzer
PerSampleAnalyzer
PerSampleAnalyzer

Testing Expanded

Sample Generation

Using a Sampling Plugin to create samples
with unique values for the 23 scalar constants

Running `elmtest.exe`

Running the program using the constants
and recording the results

Analysis

Using an Analysis Plugin to analyze the results,
especially in comparison to reference data

Constants, Inputs, and Outputs

- The 26 *constants* control how the model behaves and manipulates data.
  - We don’t yet support modification of the 3 array constants. Of the other 23 scalar constants we do experiment with, 8 have no observable effect on the output.
- The 25 *inputs* provide starting data for the model.
- The 9 *outputs* constitute data that the model has generated.
- 8 variables are both an input and an output.

Table of Constants

Array Constants	crop, iscft, percrop
Scalar Constants with Observable Outputs	Betavis, cnfac, cpliq, denh2o, grav, hfus, lake_no_ed, lakepuddling, nlevgrnd, nlevlak, nlevsno, tkwat, vkc
Scalar Constants without Observable Outputs	cpice, denice, depthcrit, mixfact, pudz, tkice

Tfrz, tdmax, n2min, and za_lake from LakeTemperature.md not included yet, don't know if their outputs are observable or not

This table uses explicitly the constants mentioned in LakeTemperature.md plus the array constants

Table of Inputs and Outputs

Inputs	col_es%t_grnd, col_pp%dz, col_pp%dz_lake, col_pp%z, col_pp%z_lake, col_pp%zi, col_pp&lakedepth, col_pp&snl, col_ws%h2osno, col_ws%h2osoi_ice, col_ws%h2osoi_liq, lakestate_vars%etal_col, lakestate_vars%ks_col, lakestate_vars%lake_raw_col, lakestate_vars%ws_col, soilstate_vars%csol_col, soilstate_vars%tkmg_col, soilstate_vars%tksatu_col, soilstate_vars%watsat_col, solarabs_vars%fsds_nir_d_patch, solarabs_vars%fsds_nir_i_patch, solarabs_vars%fsr_nir_d_patch, solarabs_vars%fsr_nir_i_patch, solarabs_vars%sabg_lyr_patch, solarabs_vars%sabg_patch
Both Inputs and Outputs	col_es%t_lake, col_es%t_soisno, lakestate_vars%lake_icefrac_col, lakestate_vars%lake_icethick_col, veg_ef%eflx_gnet veg_ef%eflx_sh_grnd, veg_ef%eflx_sh_tot, veg_ef%eflx_soil_grnd
Outputs	ch4_vars%grnd_ch4_cond_col, col_ef%errsoi, col_es%hc_soi, col_es%hc_soisno, col_wf%qflx_snofrz, col_ws%frac_iceold, lakestate_vars%betaprime_col, lakestate_vars%lakeresist_col, lakestate_vars%savedtke1_col

NetCDF

- Network Common Data Form (*NetCDF*) is a format for self-describing datasets. Each file is composed of “dimensions” and “variables.” Variables are multi-dimensional arrays, with the meaning of each dimension described in the file.

```
netcdf x_times_y {
dimensions:
        x = 5 ;
        y = 5 ;
variables:
        float x(x) ;
        float y(y) ;
        float z(x, y) ;
data:

 x = -2, -1, 0, 1, 2 ;

 y = -4, -2, 0, 2, 4 ;

 z =
  8, 4, -0, -4, -8,
  4, 2, -0, -2, -4,
  -0, -0, 0, 0, 0,
  -4, -2, 0, 2, 4,
  -8, -4, 0, 4, 8 ;
}
```

What is a Sample?

- A sample is a set of values for the constants contained in `spel-constants0001.nc`.
- `elmtest.exe` is run once for each sample.

What is a Sample? (Illustration)

Constants (formerly lakeparams.txt)

```
betavis
0.0
za_lake
0.6
n2min
7.4999999999999993E-005
tdmax
277.0
pudz
0.0
mixfact
10.0
depthcrit
25.0
```

Example Sample (JSON used for illustration)


```
{
  "betavis": 0.0,
  "za_lake": 0.6,
  "n2min": 7.4999999999999993E-005,
  "tdmax": 277.0,
  "pudz": 0.0,
  "mixfact": 10.0,
  "depthcrit": 25.0
}
```

What Outputs are Received per Sample?

- Analysis plugins will receive the outputs from the `spel-outputs0001.nc` file in the form of multidimensional `numpy` arrays, which range from one to three dimensions.

Outputs Received by Analysis Plugins (JSON for illustration)

```
{
  "veg_ef%eflx_soil_grnd": [0.0, ...],
  "veg_ef%eflx_gnet": [[0.0, ...], ...],
  ...
}
```

Sample Groups

- Represent a grouping of ordered samples
  - When the program is run on each sample in a sample group, the model data is returned in the same order as the samples were in.
- Analysis plugins can analyze the result of multiple consecutive runs of `elmtest.exe`.
- Having multiple sample groups is useful because each one can cover a range of values for a specific combination of constants whose combined effect is being examined.

Sample Group Illustration

Sample Group Data (JSON for illustration)

```
[
  {
    "betavis": 0.0,
    ...
  },
  {
    "betavis": 0.25,
    ...
  },
  {
    "betavis": 0.5,
    ...
  }
]
```

Ordered Output Data (JSON for illustration)

```
[
  {
    "lakestate_vars%betaprime_col": [0.0, ...],
    ...
  },
  {
    "lakestate_vars%betaprime_col": [0.25, ...],
    ...
  },
  {
    "lakestate_vars%betaprime_col": [0.5, ...],
    ...
  }
]
```

Plugins

- Our testing framework supports two types of plugins, sampling and analysis. Sampling plugins define values for constants, and analysis plugins view the outputs of the model.
- Two sampling plugins are currently supported:
  - CSV Index Sampling, Order Sampling
- Four analysis plugins are currently supported:
  - Change Detection, Differences, Fault Finding, NetCDF4 Output

Sampling Plugin Overview and Visuals

- Sampling plugins define a subclass of `Sampler`, returning one or more `SampleGroups`.

Interface

```python
class Sampler(ABC):
    @abstractmethod
    def get_sample_groups(self) -> Mapping[str, SampleGroup]:
        pass
```

UML Diagram

Current Sampling Plugins

CSV Index Sampling	Applies combinatorial testing using CSV files generated by NIST's ACTS (short, high level description) Uses CSV files generated by *ACTS* (Automated Combinatorial Testing for Software)[1] to create samples. It will take the largest index and use that to linearly interpolate between the values for each constant found in the ranges file
Order Sampling	Uses JSON files called "orders" to create "lines" in the input space (the constants), through linear interpolation

1. "Automated Combinatorial Testing for Software (ACTS)." *National Institute of Standards and Technology*, 20 July 2016, csrc.nist.rip/groups/SNS/acts/.

Analysis Plugin Overview

- Analysis plugins can either be “per sample” or “sample group” analysis plugins.
- “Per sample” plugins define a subclass of `PerSampleAnalyzer`, whose method is called each time `elmtest.exe` is run, and which receives the outputs produced by that specific sample.
- “Sample group” plugins define a subclass of `SampleGroupAnalyzer`, whose method is called after all samples in a group are run through `elmtest.exe`, and which receives all outputs from each.

Analysis Plugin Visuals

Interfaces

```python
class PerSampleAnalyzer(ABC):
    @abstractmethod
    def analyze_sample_data(
        self,
        sample: Sample,
        reference_data: ModelData,
        test_data: ModelData,
    ) -> tuple[str, FileSystemTree]:
        pass
```

```python
class SampleGroupAnalyzer(ABC):
    @abstractmethod
    def analyze_sample_data(
        self,
        sample_group: SampleGroup,
        reference_data: ModelData,
        sample_data: list[ModelData],
    ) -> tuple[str, FileSystemTree]:
        pass
```

UML Diagram

Current Analysis Plugins

Per Sample Analyzers	Change Detection	Identifies, per sample group, which inputs and outputs differ at some point from the reference
	Differences	Finds differences between values in the reference data and the current sample's data. It records the indices of the difference, the two differing values, and the difference between them.
	NetCDF4 Output	Stores each sample group as an individual NetCDF file, to be used for graphing changes based on our samples. To do this, it just adds a new dimension "sample_index" and layers each sample's NetCDF file on top of the others using that.
Sample Group Analyzer	Fault Finding	Uses "check" functions defined by the plugin user to make assertions about values in the sample data, similar to a unit testing framework but with assertions pertaining to model values rather than specific function behavior

Configuration